

TNSDC – IBM AGENTIC AI PROJECT

AI E-Commerce Customer Support Agent

An Agentic AI system for personalized learning, question answering, and quiz generation

Project Type	Agentic AI Mini Project
Selected Use Case	Use Case 3– AI E-Commerce Customer Support Agent
Core Capabilities	Memory + Tool calling
Development Duration	5 -day project

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1. Project Overview

1.1 Project Title

AI E-Commerce Customer Support Agent

1.2 Problem Statement

E-commerce platforms receive a large number of customer queries related to orders, products, payments, delivery, returns, refunds, cancellations, and account-related issues. Handling these queries manually can be time-consuming and may result in delayed responses.

Traditional customer-support systems often depend on predefined rules or FAQs. They may not understand the customer's complete request or provide personalized responses.

Therefore, there is a need for an AI-powered customer support agent that can understand customer queries, identify their intent, retrieve relevant information, and provide accurate responses automatically.

1.3 Brief Description of the Project

The AI E-Commerce Customer Support Agent is an Agentic AI-based system designed to assist customers with common e-commerce queries.

The agent uses natural-language processing and AI reasoning to understand customer requests. Depending on the query, it can provide information about orders, delivery, returns, refunds, products, and other support-related topics.

The system aims to reduce the workload of human support teams while providing customers with fast, consistent, and helpful assistance.

2. Objectives & Proposed Solution

2.1 Project Objectives

The main objectives of the project are:

- To develop an AI-based customer-support agent for e-commerce platforms.

- To understand customer questions written in natural language.

- To provide quick and relevant responses to customer queries.

- To automate frequently requested customer-support tasks.

- To reduce the workload of human customer-support representatives.

- To improve customer experience through 24/7 assistance.

- To provide personalized responses based on the customer's request and available information.

- To create a system that can be extended with additional tools and services in the future.

2.2 How the Agentic AI Solution Works

The system follows an agent-based workflow:

Customer Query → Query Understanding → Intent Identification → Information Retrieval/Tool Use → AI Reasoning → Response Generation → Customer

Step 1: Customer submits a query

The customer enters a question or request in the chat interface.

Example:

"Where is my order?"

Step 2: Query understanding

The AI agent analyzes the customer's message and identifies what the customer is asking.

For example, the system recognizes that the query is related to order tracking.

Step 3: Intent identification

The agent determines the appropriate category of the request, such as:

Order tracking
Order cancellation
Return request
Refund status
Product information
Payment issue
Delivery issue
General FAQ

Step 4: Information retrieval or tool usage

If information is required, the agent can use an appropriate knowledge source, database, API, or tool. For example, for an order-tracking request, the agent can retrieve the relevant order information.

Step 5: AI reasoning

The agent processes the retrieved information and determines the appropriate response.

Step 6: Response generation

The AI generates a natural-language response that is easy for the customer to understand.

Step 7: Human escalation

If the request is complex or cannot be resolved automatically, the system can recommend escalation to a human customer-support representative.

2.3 Key Features

1. Natural Language Interaction

Customers can communicate with the system using normal conversational language.

2. Order Support

The agent can assist with queries related to order status, order details, and delivery.

3. Return and Refund Assistance

The agent can provide information about return procedures and refund-related queries.

4. Product Information

Customers can ask questions about available products and their features.

5. FAQ Assistance

The agent can answer frequently asked questions using a predefined knowledge base.

6. Intelligent Query Routing

The system identifies the type of request and determines the appropriate action.

7. Personalized Responses

Responses can be generated according to the customer's specific query and available information.

8. 24/7 Availability

An AI agent can provide support outside normal customer-service working hours.

9. Human Escalation

Complex or unresolved problems can be transferred to human support.

3. Implementation & Results

3.1 Technologies/Tools Used

The project can be implemented using the following technologies:

Technology/Tool

Purpose

IBM watsonx.ai

AI model development and experimentation

IBM watsonx Assistant

Conversational AI and customer interaction

Python

Backend logic and AI integration

Generative AI / LLM

Understanding queries and generating responses

Knowledge Base

Storing FAQs and support information

Database

Storing customer/order-related information

API Integration

Connecting the agent with external services

Web Interface

Providing the customer chat interface

The exact tools used should be changed to match the tools you actually used during your internship/project.

3.2 Working Process

The working process of the proposed system is:

Step 1: User Interaction

The customer opens the e-commerce support chatbot and enters a query.

Step 2: AI Analysis

The agent analyzes the customer's input and determines the purpose of the request.

Step 3: Context Understanding

The agent considers the conversation context so that customers do not need to repeat information unnecessarily.

Step 4: Knowledge Retrieval

The system retrieves relevant information from the knowledge base or connected data source.

Step 5: Decision Making

The agent determines whether it can answer the question directly or needs to perform an action/use a tool.

Step 6: Response

The agent provides the customer with a clear and relevant answer.

Step 7: Escalation

If the agent cannot resolve the issue, it can direct the customer to human support.

Example

Customer:

"I want to know whether my order has been shipped."

Agent:

The system identifies the request as an order-status query, retrieves the relevant order information, and provides the current status to the customer.

4. Screenshots/Output

This section should contain screenshots demonstrating the working system.

Screenshot 1 – Home/Chat Interface

Show the main customer-support chatbot interface.

Caption:

Figure 1: AI E-Commerce Customer Support Agent interface

Screenshot 2 – Customer Query

Show a customer entering an e-commerce-related question.

Caption:

Figure 2: Customer submitting a support query

Screenshot 3 – AI Response

Show the AI agent generating a response.

Caption:

Figure 3: AI-generated response to the customer query

Screenshot 4 – Order/Return Support

Show an example of the agent handling an order, return, refund, or delivery-related query.

Caption:

Figure 4: Agent handling an e-commerce support request

Screenshot 5 – Final Output

```
PS C:\Users\DEEPAKKUMAR\AppData\Local\Packages\5319275A.WhatsAppDesktop_cv1g1gvanyjgm\LocalState\sessions\0713962f681fba0ed98580a3219c8eb4fd7d14e\transfers\2026-36> & 'c:\Users\DEEPAKKUMAR\AppData\Local\Packages\5319275A.WhatsAppDesktop_cv1g1gvanyjgm\LocalState\sessions\0713962f681fba0ed98580a3219c8eb4fd7d14e\transfers\2026-36\demo_run.py'

AI E-COMMERCE SUPPORT AGENT – FULL EXECUTION TRANSCRIPT
(model calls scripted for offline/no-network demo; all tool
execution, order/product lookups, and memory read/writes are
real code from ecommerce_support_agent.py)
=====

[SYSTEM] connecting as Jordan Lee (CUST-001), loyalty tier: Gold

You: Where is my order ORD-10001?

Aria: Your order ORD-10001 (AeroFlex Running Shoes) has shipped via UPS and is on its way – tracking number 12999AA10123456784. Estimated delivery is tomorrow.

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You: Can you recommend something for hiking under $150?

Aria: I'd recommend the Trailblazer Hiking Boots ($129.99, 4.7★) – waterproof with reinforced ankle support, well within your $150 budget. Want me to pull up more details or check sizing?

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You: I want to return the headphones from ORD-10002, they arrived defective

Aria: I've started your return for the CloudComfort Wireless Headphones from ORD-10002 – reason logged as 'arrived defective'. Refund amount: $149.99. You'll get an email with a prepaid shipping label shortly. Is there anything else I can help with?

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You: What's the status of that return?

Aria: Your return is currently in 'initiated' status – we're waiting on the item to reach our warehouse. Once it's received and inspected, your $149.99 refund will be processed to your original payment method.

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You: By the way, I usually wear a size 9 and prefer eco-friendly packaging

Aria: Got it – noted that you're a size 9 and prefer eco-friendly packaging. I'll keep that in mind for future recommendations and orders.

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[SYSTEM] Session ended. Long-term memory persisted to:
c:\Users\DEEPAKKUMAR\AppData\Local\Packages\5319275A.WhatsAppDesktop_cv1g1gvanyjgm\LocalState\sessions\0713962f681fba0ed98580a3219c8eb4fd7d14e\transfers\2026-36\memory_store\CUST-001.json
```

Results Achieved

- The proposed AI customer-support agent provides several benefits:
- Customers can interact with the system using natural language.
- Common customer queries can be handled automatically.
- Responses can be generated quickly.
- Frequently asked questions can be answered using a knowledge base.
- Customer-support workload can be reduced.
- The system can provide assistance continuously.
- The agent can identify different types of e-commerce requests.
- Complex issues can be directed to human support.
- The architecture can be extended with additional tools and APIs.

Sample Result

The developed system successfully demonstrates how Agentic AI can be applied to e-commerce customer support. The agent can understand customer requests, retrieve relevant information, reason about the request, and generate an appropriate response.

6. Conclusion & Future Scope

6.1 Project Conclusion

- The AI E-Commerce Customer Support Agent demonstrates the potential of Agentic AI in improving customer-support operations.
- Instead of relying entirely on manual customer service, an AI agent can understand customer requests and provide automated assistance for common e-commerce problems.
- The project demonstrates how AI can be combined with knowledge bases, APIs, databases, and conversational interfaces to create a more intelligent customer-support solution.
- Overall, the system can help improve response time, customer convenience, and support-team efficiency.

6.2 Challenges Faced

During development, some potential challenges include:

- Understanding different ways customers phrase the same question.
- Maintaining context across multiple messages.
- Providing accurate information.
- Handling ambiguous customer queries.
- Connecting the AI agent with external databases or APIs.
- Managing incorrect or incomplete customer information.
- Deciding when a query should be transferred to human support.
- Ensuring that AI-generated responses remain relevant and consistent.

6.3 Future Enhancements

The project can be improved in several ways:

1. Multilingual Support

The agent could support multiple languages so that customers from different regions can interact more easily.

2. Voice-Based Support

Voice interaction could be added to allow customers to communicate with the agent through speech.

3. Advanced Personalization

The system could use customer and order context to provide more personalized assistance.

4. Real-Time Order Tracking

Integration with an e-commerce order-management system could allow the agent to provide real-time delivery information.

5. Automated Actions

The agent could perform approved actions such as initiating return requests or checking refund status.

6. Sentiment Analysis

The system could detect customer sentiment and prioritize frustrated or urgent customers for human assistance.

7. Human-Agent Collaboration

The AI agent could summarize the conversation before transferring a complex issue to a human representative.

8. Continuous Learning

The system could be improved using feedback from customer interactions and support representatives.

7. References

You can include the official documentation and learning resources that you actually used. For an IBM-based project, examples include:

IBM watsonx.ai documentation

IBM watsonx Assistant documentation

Python documentation

Documentation for the APIs/databases used in the project

Course or internship training materials provided by IBM